

Components of IoT Application

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All work for this course can be found in [this github repository](#). There I saved code snippets and other artifacts related to this course. It might be easier to read corresponding `md` files rather than a long pdf submitted for the assignment. Working on assignments for this course I initially wrote answers in my own words and before committing the changes I used Claude to verify my answers and improve wording and grammar, and to unify the formatting across all weeks.

Week 1

1. Define foundational networking terms. Write 2–3 sentences for each term. State what it means, why it matters, and, where useful, give one concrete example or relationship to another term.

Performance and protocol data:

- **network bandwidth** — a parameter that shows how much data can be transmitted.
 - Ever-growing need
 - Limited availability
 - Smart design and usage saves money
 - One important factor to estimate is "how good" the network actually is
- **network throughput** — Actual data successfully delivered over the link
- **packet loss and jitter** — during the data transmission, packets can be lost. It can be referred to as **packet loss**. Even a packet loss rate of 5% can severely reduce effective throughput, depending on the network conditions and communication protocol.
Jitter describes the variation in packet delay or round-trip time (RTT). High jitter can lead to interruptions and reduced performance in communication protocols, especially for real-time applications.

- **bps versus Bps:** *bps* — bits per second (b/s), *Bps* — byte per second (B/s). 1 Bps = 8 bps.

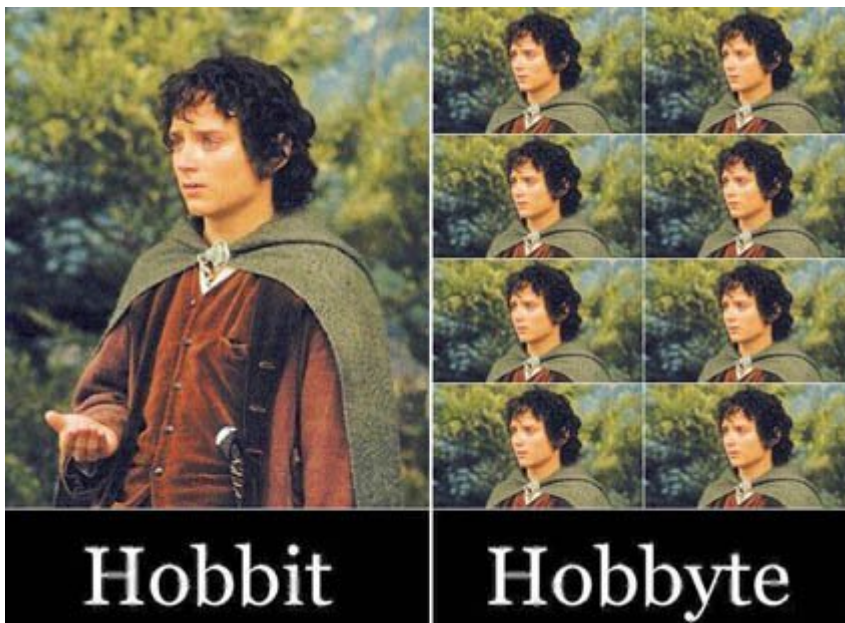


Figure 1.1 — Hobbit/hobbyte joke. Credit to unknown reddit user

- **protocol payload** — the actual useful data a packet carries, for example the sensor reading or the file contents, without any headers. Every protocol layer treats the whole packet from the layer above as its payload. For example, a TCP segment is the payload of an IP packet.
- **protocol overhead, especially in resource-constrained IoT systems** — everything that has to be sent besides the payload: headers, trailers, handshakes, acknowledgements and retransmissions. For a 4-byte temperature reading sent over TCP/IPv6/Ethernet, the headers alone are about 80 bytes, so most of the transmitted data is overhead. On battery-powered IoT devices with slow radios, every extra byte costs energy and airtime. That's why IoT uses compact protocols like MQTT, CoAP over UDP and 6LoWPAN header compression.

Link layer, topology, media, and wireless networking:

- **Spanning Tree Protocol (STP)** — A Layer 2 protocol that prevents switching loops in Ethernet networks by blocking redundant paths. If an active link fails, a blocked path can be activated.
- **collision domain** — A network segment where devices share the same transmission medium and simultaneous transmissions can cause collisions. Switches reduce collision domains by giving each port its own domain.
- **broadcast domain** — A group of devices that receive the same Layer 2 broadcast traffic. Routers separate broadcast domains.
- **SOHO network** — A Small Office/Home Office network designed for a small number of users and devices. It typically consists of a router, Wi-Fi access point, and computers.
- **MAC (physical) address** — A unique Layer 2 hardware address assigned to a network interface. It is used to identify devices within a local network.
- **physical-layer protocol data unit (PDU)** — The PDU of the Physical Layer is a bit. The Physical Layer transmits raw bits over the transmission medium.

- **MAC-layer protocol data unit (PDU)** — The PDU of the Data Link (MAC) Layer is a frame. Frames contain MAC addresses and the encapsulated network-layer data.
- **half-duplex versus full-duplex** — Half-duplex communication allows sending or receiving, but not both at the same time. Full-duplex communication allows simultaneous transmission and reception.
- **Ethernet auto-negotiation** — A process where Ethernet devices automatically determine the highest compatible speed and duplex mode. This occurs when a connection is established.
- **hidden-node problem in wireless networking** — In wireless networks, two devices may be unable to hear each other but can both communicate with an access point. This can lead to collisions and reduced performance.
- **physical versus logical network topology** — Physical topology describes how devices are physically connected by cables or wireless links. Logical topology describes how data flows between devices.
- **TIA/EIA-568 and ISO/IEC 11801** — These are cabling standards that define requirements for structured network cabling, including wiring schemes, performance categories, and installation practices.
- **Ethernet cabling categories, for example Category 6** — Ethernet categories define cable performance characteristics such as bandwidth and supported data rates. Higher categories generally support higher speeds and frequencies. **Cat6** specs: max frequency: 250MHz, typical speed 1Gbps up to 100m, supports 10 Gbps for shorter distances (about 55m)
- **8P8C connector, often informally called RJ45** — An 8-position, 8-contact connector commonly used for Ethernet cables. Although often called RJ45, the technically correct term is 8P8C.
- **Wi-Fi ad hoc mode** — A wireless networking mode where devices communicate directly with each other without an access point. It creates a peer-to-peer network.
- **IEEE 802.11ac, 802.11ax, and 802.11be.** — These are Wi-Fi standards corresponding to Wi-Fi 5, Wi-Fi 6/6E, and Wi-Fi 7 respectively. Each generation improves speed, efficiency, and support for multiple devices.

Tracking the data path

In Windows we can track the connection with `tracert` command. Following shows path from my laptop connected to `Eduroam` network to Google public DNS server. It goes from the local network (everything with a small delay) to Vaasa - Turku - Stockholm

```
tracert 8.8.8.8
```

```
C:\git\studies\componentes_IOT_app main # ?1 ~1 tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

 1      2 ms      2 ms      2 ms    130.231.160.1
 2      2 ms      2 ms      2 ms    at-ext-bb.oulu.fi [193.167.221.66]
 3      3 ms      3 ms      3 ms    oulu-r1.oulu.fi [193.167.221.3]
 4      2 ms      2 ms      3 ms    oulu1-et-0-1-2-2.ip.funet.fi [193.167.244.240]
 5     15 ms     15 ms     15 ms    oulu4.ip.funet.fi [86.50.255.117]
 6     15 ms     14 ms     14 ms    vaasa4-et-0-1-0-1.ip.funet.fi [86.50.254.230]
 7     15 ms     14 ms     15 ms    turku4-et-0-1-0-1.ip.funet.fi [86.50.254.236]
 8     15 ms     15 ms     14 ms    stockholm4-et-0-0-0-1.ip.funet.fi [86.50.254.251]
 9     15 ms     15 ms     17 ms    se-sthb1.nordu.net [109.105.102.0]
10     14 ms     14 ms     14 ms    se-bma2.nordu.net [109.105.101.93]
11      *        *        *      Request timed out.
12     15 ms     15 ms     15 ms    72.14.237.55
13     16 ms     16 ms     16 ms    209.85.242.83
14     30 ms     14 ms     15 ms    dns.google [8.8.8.8]

Trace complete.
```

Figure 1.2 — Result of `tracert` command.

2. Estimate how long does it take to download 3 TB file from cloud based backup service if network download throughput is 200 Mbps for actual payload (i.e. data)?

$\text{file_size} = 3 \text{ TB} = 3,000 \text{ GB} = 3,000,000 \text{ MB} = 24,000,000 \text{ Mb}$

$\text{download_speed} = 200 \text{ Mbps}$

$\text{download_time} = \text{file_size} / \text{download_speed} = 24,000,000 / 200 \text{ s} = 120,000 \text{ s}$

$1 \text{ h} = 3600 \text{ s}$

$\text{download_time_hours} = 33.33 \text{ h}$

3. Locate the MAC address of your mobile phone, laptop wifi interface or some other networked IT device

I have checked MAC addresses of my laptop using following command:

```
# on Windows  
ipconfig /all
```

On my phone I have checked the MAC address from the Wi-Fi settings.

I also verified my findings from my Wi-Fi router's control panel (DHCP Clients List). There you can see all devices currently connected to the network, their `Client name`, `Mac Address`, `Assigned IP`, and `Lease time`. To my surprise, the MAC address of my phone was different from the one I found in the settings. Apparently, it does some MAC randomization.

I won't add a screenshot for privacy reasons, but here are masked MAC addresses of my devices:

- Laptop: 88:D8:2E:00:00:00
- Phone:
 - 7C:7B:BF:00:00:00- actual address from phone settings
 - D6:8F:74:00:00:00 - from the router

Using [this](#) tool I have searched for OUI of my devices. Laptop showed as `Intel Corporate`, actual phone address `Samsung Electronics Co., Ltd`, and the random address that my phone actually used to connect to the router had no matches in this database.

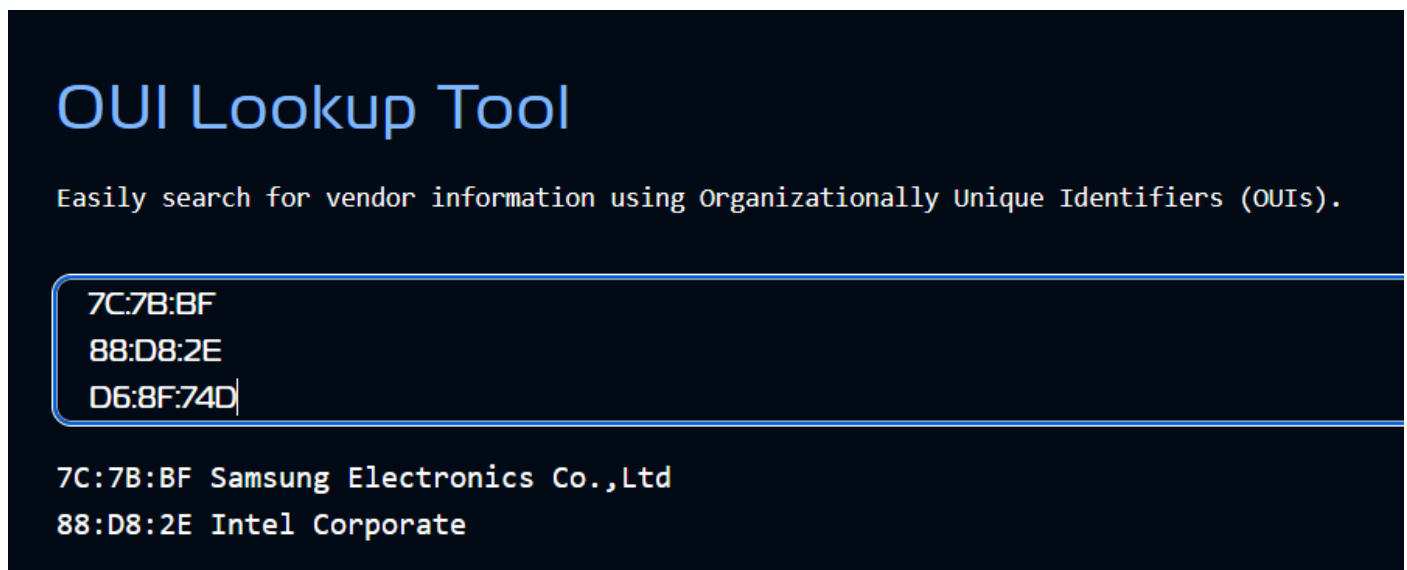


Figure 1.3 — QUI Lookup tool

4. Describe shortly what are these network devices, functions, and services:

- **Repeater** — Receives a signal and retransmits it to extend the communication distance.
- **Hub (multiport repeater)** — Connects multiple devices and forwards incoming data to all ports, regardless of the destination.
- **Bridge** — Connects two network segments and forwards traffic only when necessary based on MAC addresses.
- **Access switch** — Connects end devices such as PCs, printers, and access points to the local network.
- **Core switch** — A high-performance switch that interconnects access switches and carries large amounts of network traffic.
- **Edge router** — Connects an organization's network to external networks, such as the Internet.
- **Core router** — Routes traffic between major networks within a service provider or large enterprise backbone.
- **Firewall** — Monitors and filters network traffic according to security rules to block unauthorized access.
- **Wifi AP** — Provides wireless network access and connects Wi-Fi devices to the wired network.
- **WLAN AP controller** — Centrally manages multiple wireless access points, including configuration, security, and roaming.
- **Network TAP** — A monitoring device that copies network traffic for analysis without affecting the original traffic flow.

5. RFC assignments

- What are RFCs?

RFCs (Request for Comments) are technical documents that describe how Internet technologies, protocols, procedures, and standards work. They are published by organizations such as the **Internet Engineering Task Force (IETF)** and serve as the official reference for many Internet standards.

Historically, RFCs started as informal documents shared among researchers working on the early ARPANET. Despite the name, many RFCs today are official Internet standards.

- How many PPP related RFC documents can you find from rfc-editor website?

The screenshot shows the RFC Editor search interface. The search bar contains 'ppp' and shows 126 results. The filters on the left include 'Search only Metadata and Abstract' (unchecked), 'Search also obsoleted / historic' (checked), and a status filter showing counts for various RFC statuses: Internet Standard (2), Best Current Practice (3), Proposed Standard (66), Draft Standard (8), Informational (38), Experimental (3), and Historic (6). The search results list 'RFC 3818: BCP 88: IANA Considerations for the Point-to-Point Protocol (PPP)' as the top result, marked as 'BEST CURRENT PRACTICE' by V. Schryver, dated June 2004.

Figure 1.4 — Search results for `ppp` on <https://www.rfc-editor.org> has 126 results.

- What is the current status of RFC1597? What is the number for updated, more recent RFC of same topic?

The screenshot shows the RFC Editor search interface with 'RFC1597' entered in the search bar, resulting in 1 result. The filters on the left show 'Search only Metadata and Abstract' (unchecked), 'Search also obsoleted / historic' (checked), and a status filter showing 1 'Informational' result. The search results list 'RFC 1597: Address Allocation for Private Internets' as the top result, marked as 'INFORMATIONAL' by Y. Rekhter, B. Moskowitz, D. Karrenberg, and G. de Groot, dated March 1994. It is noted as 'Obsoleted by RFC 1918'. The abstract states: 'This RFC describes methods to preserve IP address space by not allocating globally unique IP addresses to hosts private to an enterprise while still permitting full network layer connectivity between all hosts inside an enterprise as well as between all public hosts of different enterprises. This memo provides information for the Internet community. This memo does not specify an Internet standard of any kind.'

Figure 1.5 — RFC 1597 is Obsoleted by RFC 1918

- When was RFC5218 released?

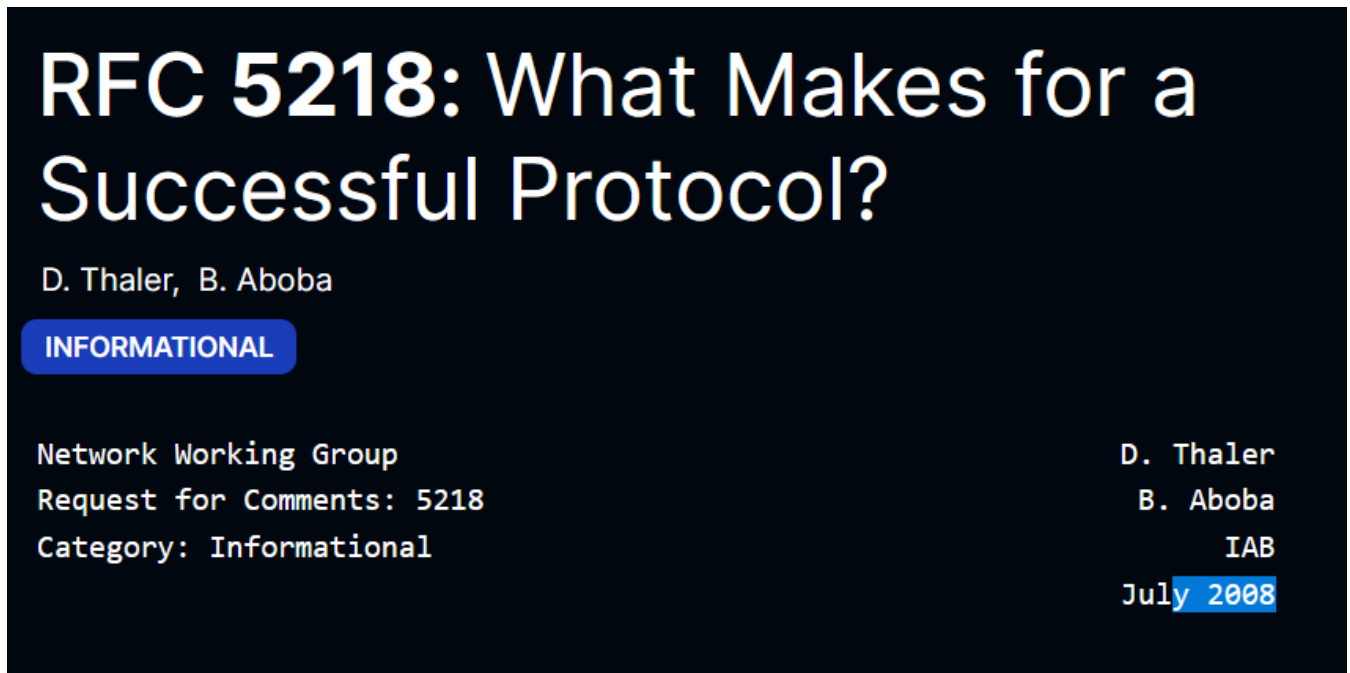


Figure 1.6 — RFC 5218 was released in July 2008

- What is the meaning if RFC status is BCP?

BCP stands for Best Current Practice.

An RFC with status BCP is not necessarily a protocol standard. Instead, it documents the recommended way to do something on the Internet based on operational experience and community consensus.

- List authors of the CoAP RFC (June 2014). What is the RFC number?

The image shows the front cover of RFC 7252. The title 'RFC 7252: The Constrained Application Protocol (CoAP)' is at the top in large white font. Below it, the authors 'Z. Shelby, K. Hartke, C. Bormann' are listed. A blue badge with the text 'PROPOSED STANDARD' is on the left. A yellow banner across the middle contains the text 'This RFC was updated, see RFC 7959, RFC 8613, RFC 8974, RFC 9175, RFC 9876.' with a right-pointing arrow. At the bottom, the Internet Engineering Task Force (IETF) information is on the left, and the authors' affiliations (ARM, K. Hartke, C. Bormann, Universitaet Bremen TZI) and the date 'June 2014' are on the right.

RFC 7252: The Constrained Application Protocol (CoAP)

Z. Shelby, K. Hartke, C. Bormann

PROPOSED STANDARD

This RFC was updated, see [RFC 7959](#), [RFC 8613](#), [RFC 8974](#), [RFC 9175](#), [RFC 9876](#). >

Internet Engineering Task Force (IETF)
Request for Comments: 7252
Category: Standards Track
ISSN: 2070-1721

Z. Shelby
ARM
K. Hartke
C. Bormann
Universitaet Bremen TZI
June 2014

Figure 1.7 — RFC 7252: The Constrained Application Protocol (CoAP)

Authors:

- Z. Shelby
 - K. Hartke
 - C. Bormann
- Twitch.tv provides IRC access to the stream chats. Which RFC defines the original Internet Relay Chat (IRC) Protocol?

The image shows the front cover of RFC 1459. The title 'RFC 1459: Internet Relay Chat Protocol' is at the top in large white font. Below it, the authors 'J. Oikarinen, D. Reed' are listed. A blue badge with the text 'EXPERIMENTAL' is on the left. A yellow banner across the middle contains the text 'This RFC was updated, see RFC 2810, RFC 2811, RFC 2812, RFC 2813, RFC 7194.' with a right-pointing arrow. At the bottom, the Network Working Group information is on the left, and the authors' names and the date 'May 1993' are on the right.

RFC 1459: Internet Relay Chat Protocol

J. Oikarinen, D. Reed

EXPERIMENTAL

This RFC was updated, see [RFC 2810](#), [RFC 2811](#), [RFC 2812](#), [RFC 2813](#), [RFC 7194](#). >

Network Working Group
Request for Comments: 1459

J. Oikarinen
D. Reed
May 1993

Figure 1.8 — RFC 1459. This is the earliest RFC I found on IRC.

6. What is OSI model? Compare OSI model to TCP/IP model

OSI model

- ▶ OSI model (Open System interconnection) by ISO (International Organization for Standardization)
- ▶ Introduced 1984
- ▶ 7 layer model to describe networking process
- ▶ Defines common terminology and helps learning

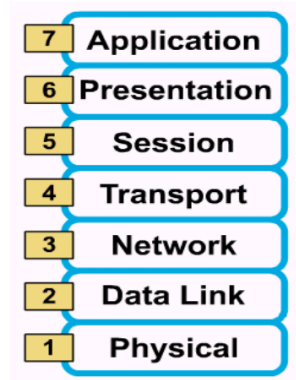


Figure 1.9 — OSI model slide

The *OSI (Open Systems Interconnection)* model is a conceptual framework that describes how data travels through a network. It divides network communication into seven layers, each with specific responsibilities.

TCP/IP model

- ▶ OPS layers 5,6 and 7 combined to Application layer
- ▶ OSI layers 1 and 2 combined to network access layer
- ▶ Sometimes used to describe IP networking

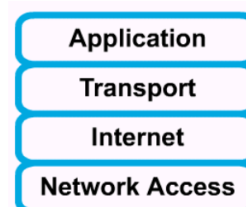


Figure 1.10 — TCP/IP model slide

TCP/IP vs OSI model

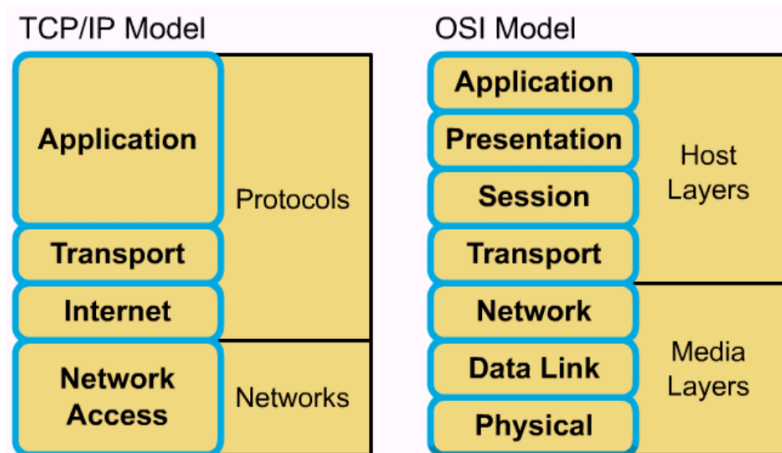


Figure 1.11 — TCP/IP vs OSI model

The **TCP/IP model** describes the same communication with only four layers, and it's the model the real Internet is built on:

OSI model	TCP/IP model	Example protocols / PDU
7 Application, 6 Presentation, 5 Session	Application	HTTP, MQTT, DNS, TLS — data
4 Transport	Transport	TCP (segment), UDP (datagram)
3 Network	Internet	IP, ICMP — packet
2 Data Link, 1 Physical	Network Access (Link)	Ethernet, Wi-Fi — frame, bits

Main differences:

- **Number of layers:** OSI has seven layers. TCP/IP has four, because it merges the top three OSI layers into *Application* and the bottom two into *Network Access*.
- **Purpose:** OSI is a theoretical reference model, designed before the protocols existed. It's mostly used for teaching and troubleshooting ("is it a layer 2 or layer 3 problem?"). TCP/IP was built from protocols that already worked, and it describes how the Internet actually runs.
- **Separation:** OSI keeps presentation (encoding, encryption) and session handling as separate layers. In TCP/IP the application handles them itself, for example TLS inside the application layer.